PC COMPONENTS

These are not the features you're looking for

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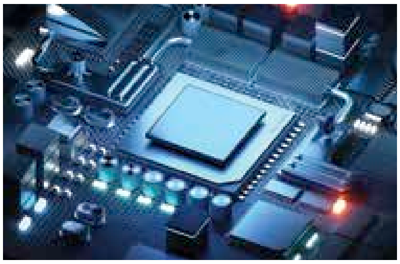
Putting together a PC is considerably easier these days given the widespread availability of eCommerce portals and abundance of choices for components. It's easier to compare prices across the length and breadth of the city before zeroing onto the components to buy. However, there's also an overabundance of gimmicky features that are shown to be the next best thing since sliced bread. We don't want you getting caught in any of the mindless marketing tricks. Which is why we're highlighting the things that you should not consider at all while buying components for your next PC build.

Processors - More is not always better

Clock speed, architecture, multi-threading and the number of cores – if your processor has an advantage in any of these areas then it's quite likely to perform



better than other less 'gifted' processors. While the architecture aspect is a little difficult to compare given that barely any information about it is made public, it's easy to gauge a better processor by looking at the other aspects ... until recently, of course. STI (Sony, Toshiba and IBM) came out with the whole heterogenous core architecture with some CPU cores being very powerful and others being more energy efficient for background tasks. This was adopted by ARM as part of their big. LITTLE architecture and also by AMD about a decade ago. Intel has also now brought out two generations of processors with a heterogeneous system architecture. So all cores within a processor stopped being equal, in terms of performance. Yet, the marketing mumbo-jumbo around these processors will still call out the total number of cores and equate that to performance. Don't fall for that. Try to count the performance-oriented cores only, that should give you a better approximation of which processor is better.

Memory - DDR5 needs more time

Memory or RAM in PCs have evolved quite a bit over the years. We're now in the era of DDR5 and a newer generation of memory has always been better for the processor, until now. With each new memory standard, there has been a doubling of raw bandwidth available for the processor. This includes all the typical operations – reading, writing and copying. What has changed now is that processors are having to play catch-up with the memory. The current generation of flagship processors can be paired with either DDR4 or DDR5 memory. The performance difference in day-to-day tasks, gaming and even content creation workloads is negligible. The higher bandwidth that DDR5 provides can only be leveraged by the processor in very niche scenarios. Let's understand this with an example. DDR4 memory clocked at 3200 MT/s provides about 33.57 Gbps of bandwidth. If you pair this with a 8-Core processor, the maximum bandwidth available for each core is 4.2 Gbps. There's not going to be any bottlenecking with As per the Steam Hardware Survey, 4-Core, 6-Core and 8-Core processors account for about 81.5 per cent of the PCs. So for the vast majority, DDR5 is not the way to go. Even if you do opt for a high core-count CPU, there are just a few tasks that can leverage the higher bandwidth.